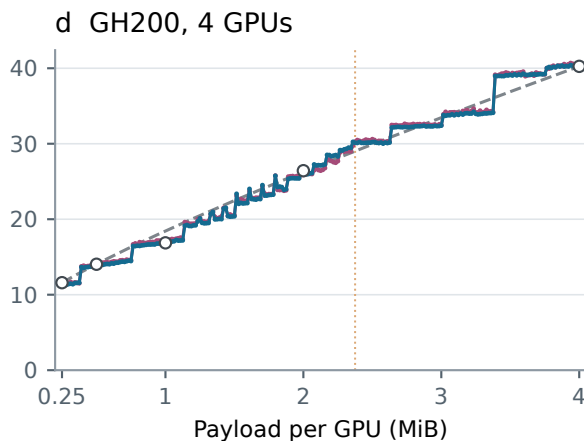
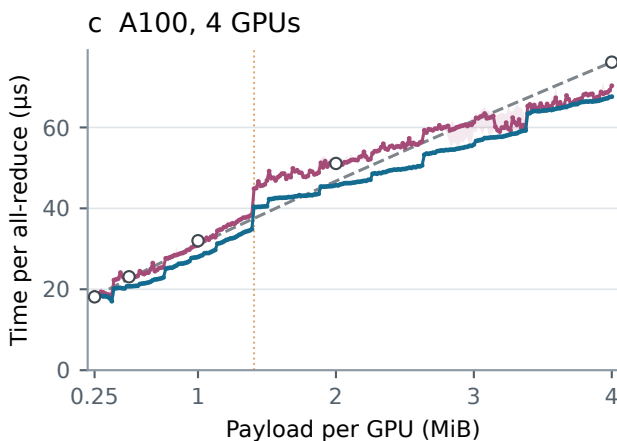
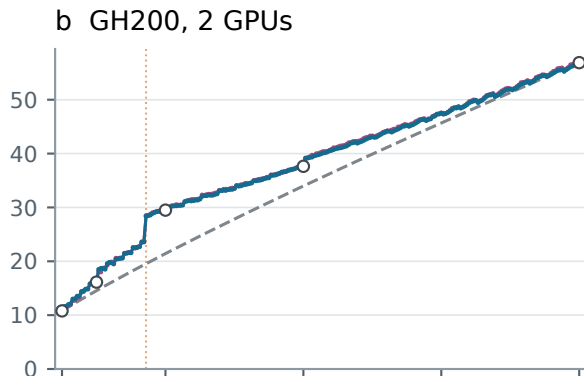
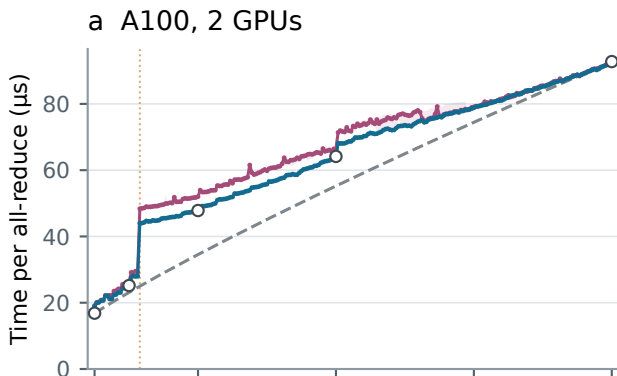


Dense all-reduce measurements around 1 MiB

Merlin | NCCL 2.31.2 | 241 sizes per curve | five process repeats

Original timing method Unchanged model
NVIDIA benchmark Earlier measurements



16 KiB spacing. Shading: repeat interquartile range. Dotted lines: protocol changes.

Timing runs have no tuning instrumentation. Model anchors and earlier data are unchanged.

Component evidence for TRAF-43; this figure does not validate end-to-end token latency.